

CONCLUSIONS

- ❖ The conceptual framework for an Arduino-based mobile robot integrating multiple control techniques is presented.
- ❖ The identification of hardware components (Arduino, sensors, motors, drivers, power supply) and software requirements (Arduino IDE, libraries, optional simulation tools) demonstrated that the design is feasible with accessible resources.



THANK YOU

APPLIED ARES

- ❖ **Robotics and Automation** – Mobile robot movement and autonomous operation.
- ❖ **Industrial Automation** – Automated material handling, inspection, and transportation.
- ❖ **Smart Manufacturing** – Intelligent control and monitoring of robotic systems.
- ❖ **Obstacle Avoidance** – Safe navigation using ultrasonic sensors.
- ❖ **Remote Control Systems** – Bluetooth and IR-based wireless robot control.
- ❖ **Voice-Controlled Systems** – Smartphone or voice-command-based robot operation.



SMART MANUFACTURING



TECHONOLOGICAL UNIVERSITY (MANDALAY)
DEPARTMENT OF MECHATRONIC ENGINEERING

Integration of Multiple Control Techniques in an Arduino-Based Mobile Robot



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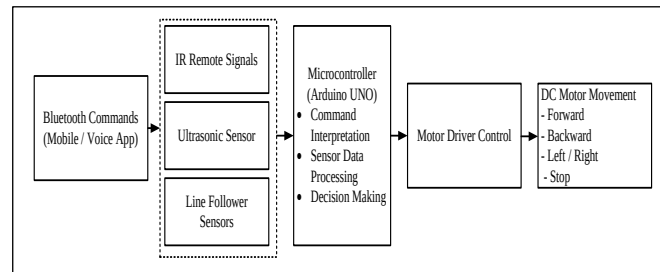
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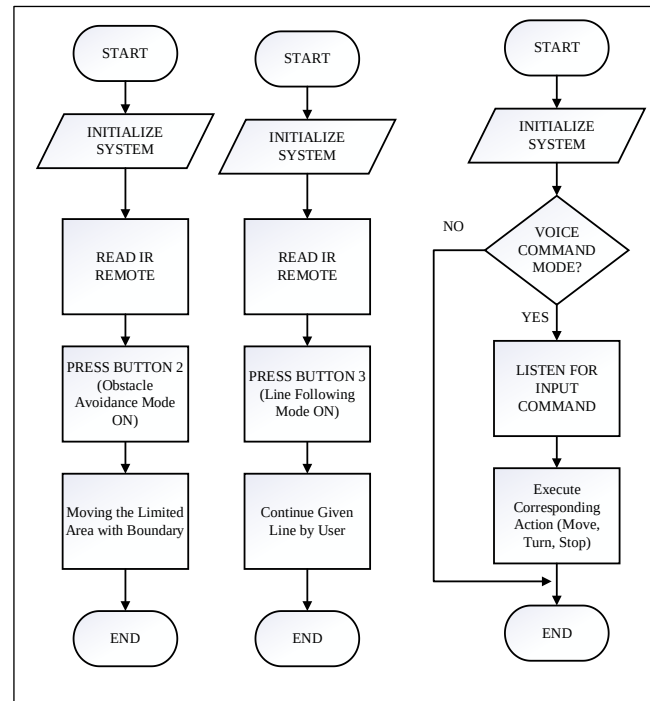
The main aim of this project is to construct a smart mobile robot that can operate in both manual and autonomous modes using different control techniques.

OBJECTIVES

- ❖ To design and develop a multifunction Arduino-based robot car that can operate in both manual and autonomous modes.
- ❖ To implement multiple control methods such as Bluetooth control, voice control, IR remote control, line following, and obstacle avoidance in a single robotic system.
- ❖ To interface and utilize different sensors and modules (ultrasonic sensor, IR sensors, Bluetooth module) for environment sensing and wireless communication.
- ❖ To enhance learning of embedded systems and robotics concepts including motor control, sensor integration, and real-time decision-making using Arduino.



Overall block diagram of this project



Flowchart of the Obstacle Avoidance/Line Following/Voice Command Mode

Hardware Components

- ❖ Arduino Microcontroller (Central Processing Unit)
- ❖ Bluetooth Module (HC-05) – Bluetooth control and voice commands
- ❖ IR Sensors – Line following and remote control
- ❖ Ultrasonic Sensor (HC-SR04) – Obstacle detection
- ❖ Motor Driver (L298N) – Controls DC motors
- ❖ DC Motors and Wheels – Movement
- ❖ Power Supply – Battery for the robot

